



W O M E N H E A L T H

Breast Cancer Diagnosis

A Statistically Validated Comparison of Machine Learning Models

ISA 640: Capstone Project | Summer 2026

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Agenda



Problem & Hypothesis

Why this matters, research questions, and our testable hypothesis



Data & Feature Engineering

Neo4j pipeline, cleaning, engineered features, EDA



Baseline & Model Comparison

Justified baseline, six model families, hyperparameter tuning



Statistical Validation & Conclusion

Significance testing, error analysis, final takeaways

Why This Matters?

- **Breast cancer (BC)** is the most common cancer among women, making it essential to have an accurate and dependable system for diagnosing benign or malignant tumors.
- **1 in 40 women will have BC, 1 in 8 women diagnosed with BC will die** from this condition (WHO).
- **Effective treatment** depends on early detection.
- The use of ML and AI to develop precise and effective BC detection and classification methods has gain increasing attention.

BC in Women Worldwide (Incidence Vs. Death Rates)

Table 1. BC incidence rates in women worldwide: source [13].

Rank	Country	Number	ASR/100,000
	World	2,261,419	47.8
1	Belgium	11,734	113.2
2	The Netherlands	15,725	100.9
3	Luxembourg	497	99.8
4	France	58,083	99.1
5	France, New Caledonia	185	99.0
6	Denmark	5083	98.4
7	Australia	19,617	96.0
8	New Zealand	3660	93.0
9	Finland	5228	92.4
10	US	253,465	90.3



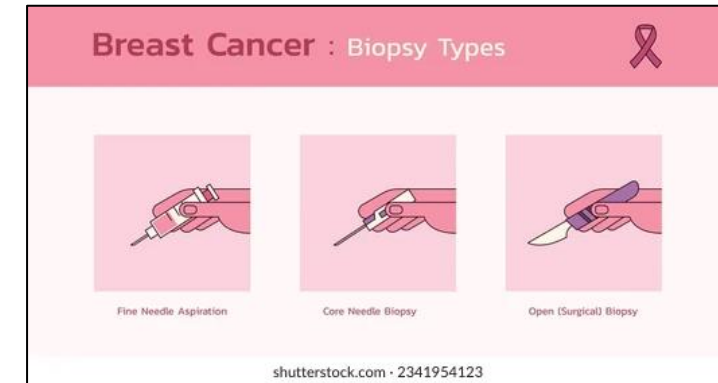
Table 2. BC death rates in women worldwide: source [13].

Rank	Country	Number	ASR/100,000
	World	684,996	13.6
1	Barbados	111	42.2
2	Fiji	184	41.0
3	Jamaica	637	34.1
4	Bahamas	80	31.0
5	Papua New Guinea	847	27.7
6	Somalia	1189	27.2
7	Mali	1425	26.6
8	Dominican Republic	1577	26.4
9	Syria	1946	26.2
10	Samoa	21	25.6

Project Overview

This study:

- Utilizes fine needle aspiration (FNA) cytology to enhance clinical capacity.
- In Biopsy, breast tumors measurements derived from these image techniques are utilized to distinguish between malignant and benign tumors
- Aims to **detect** and divide BC into 2 categories (M: Malignant, B: Benign)
- Explores automated BC prediction using:
 - Multi-model features, which incorporates: LR, DT, RF, SVM, KNN, GBT
 - Ensemble machine learning (EML), which incorporates voting, bagging, staking, and boosting
 - MLP (Multi-layer perceptron)
- Furthermore, I conducted:
 - Train/test strategy: 80/20 (455 train / 114 test)
 - Cross validation experimentation
 - Six evaluation matrix, including: precision, sensitivity, AUC, specificity, accuracy, and F-1 scores



Research Questions & Hypothesis

- How accurately can supervised ML models predict malignant vs. benign diagnosis from **FNA-derived diagnostic features**?
- Does a more complex, non-linear model (Random Forest) provide a **statistically significant** improvement over a simpler, interpretable linear baseline (Logistic Regression)?

H1:

- Ensemble-based and margin-based classifiers (Random Forest, SVM) will **outperform** simpler linear models (Logistic Regression), given expected non-linear relationships among cell-nucleus measurements.

H1 is testable and falsifiable: it predicts a statistically significant accuracy gain for the best complex model over the tuned baseline

We will fail to accept this as we will see in the next slides.

1- Dataset Acquisition and Pre-processing

- In this study the **Wisconsin Diagnostic Breast Cancer (WDBC)** dataset was used for the model training.
- The **University of California Irvine (UCI)** ML repository provided this dataset.
- It contains 30 tumors characteristics from 569 patients.
- The **features are calculated** from a digital image of a breast mass that is aspirated with an FNA.
- The 30 features include an **instance ID**, a class label indicating whether each instance has a benign or malignant tumor, and 30 real tumors traits.



* Reference: <https://archive.ics.uci.edu>

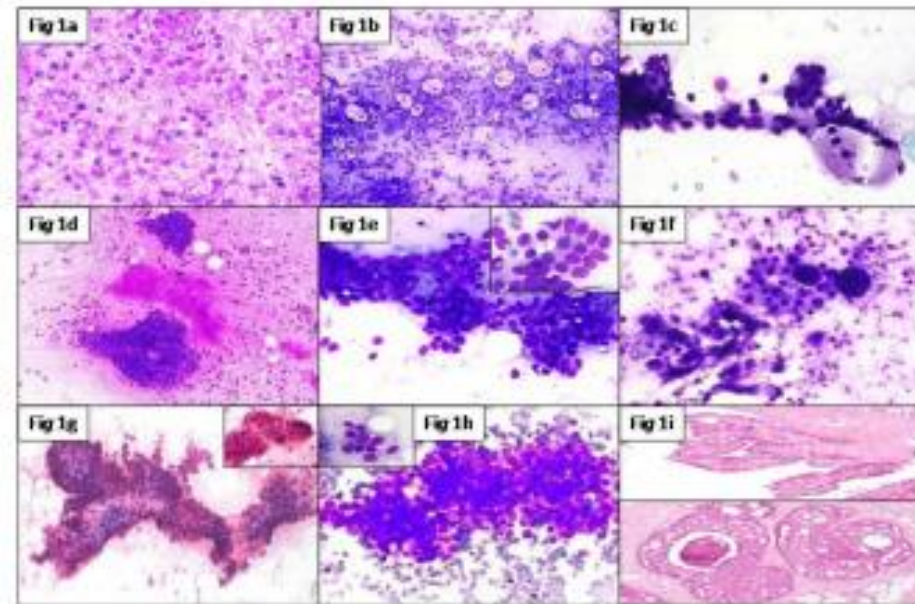
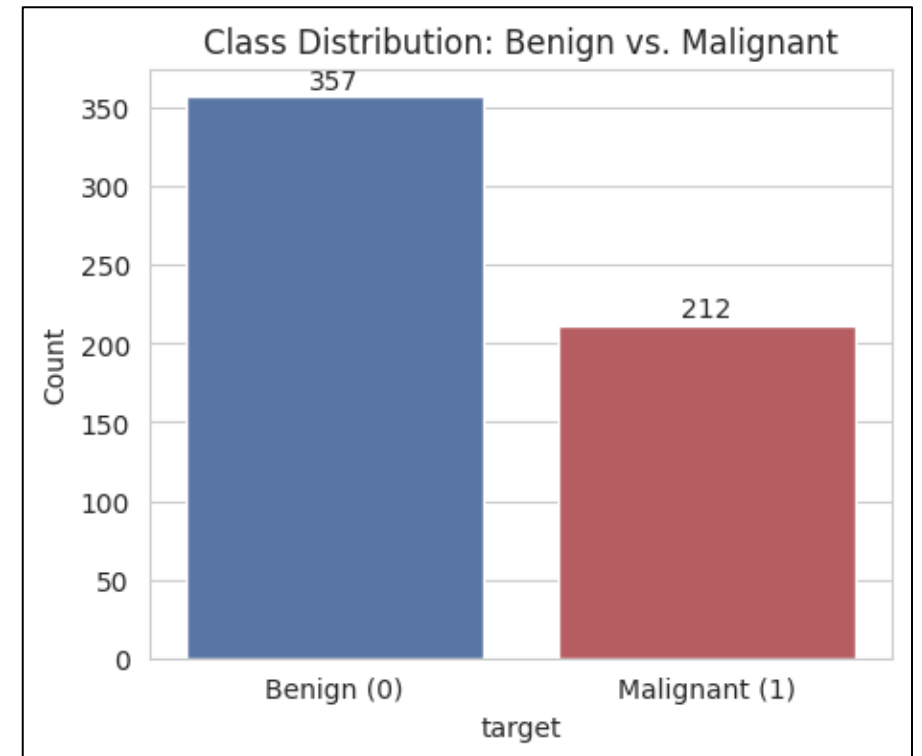


Fig1a & 1b : Breast abscess, Fig1c : Fibrocystic changes, Fig1d&1e : Fibroadenoma (Benign)
Fig1f, 1g & 1h: Atypical cells , Fig1i: DCIS

EDA of the WDBC Dataset

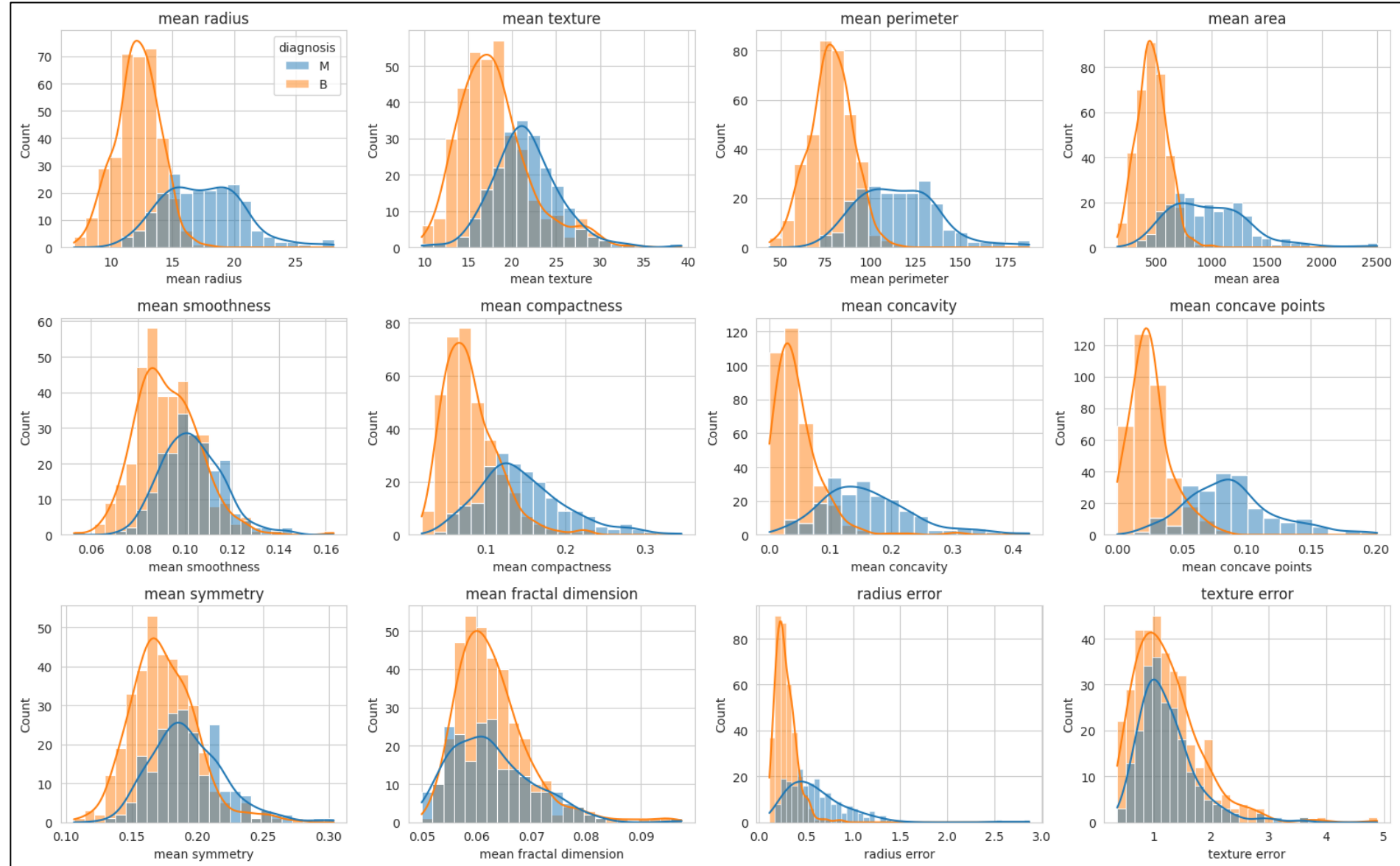
- There are 357 (~62%) instances of benign, and 212 (~37.3%) malignant tumors.
- **No resampling** is done; the data was mildly distributed.
- The **exploratory analysis** presents:
 - Histograms plots for the distribution of features in the dataset for the two tumor classes (0: benign and 1: malignant)
 - Heatmap, to determine correlations among different features in BC data
 - Correlation with the target class



Histogram Plots for the Features Distribution by Class

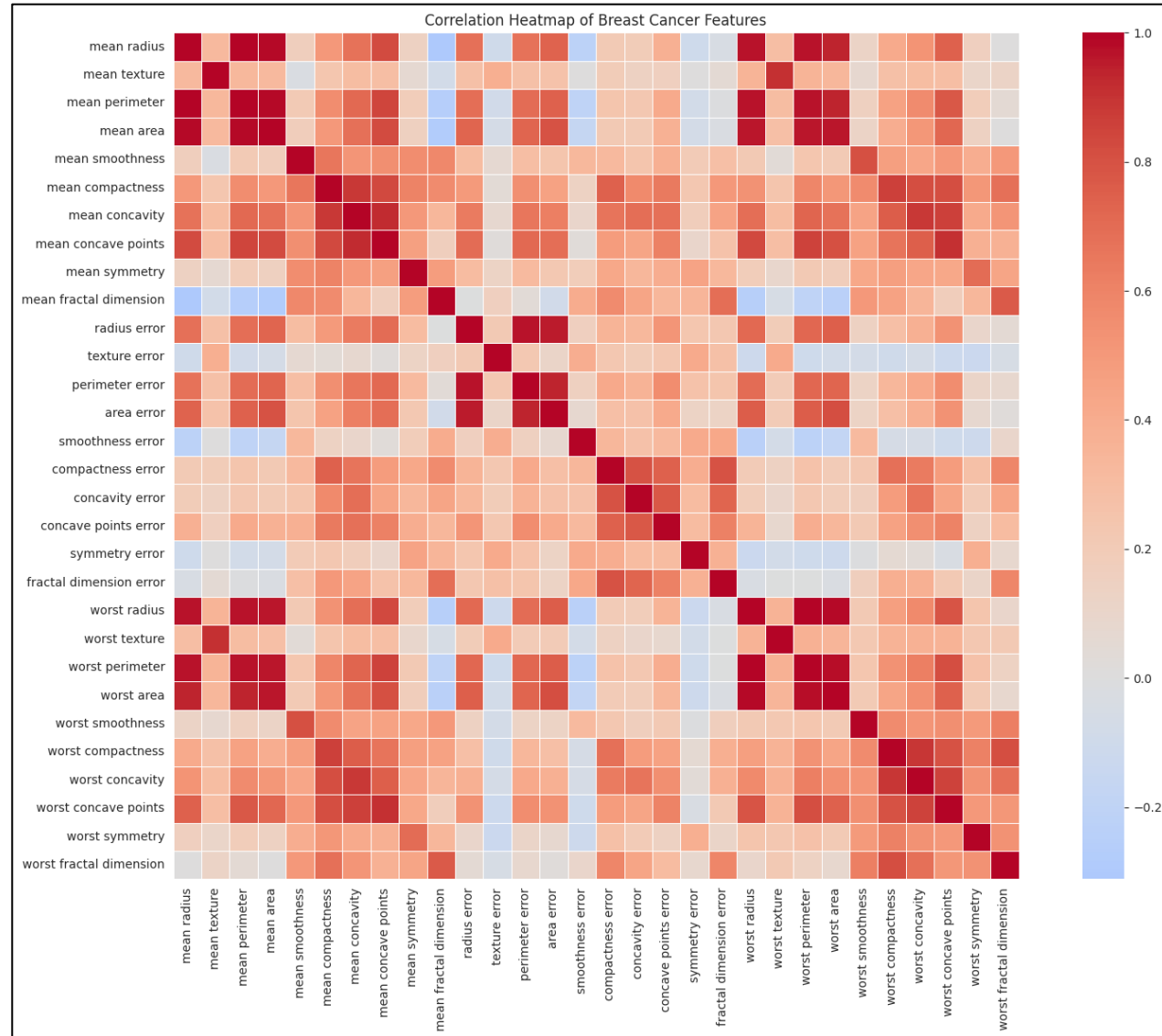
- **Size and shape-irregularity features** clearly separates the classes:

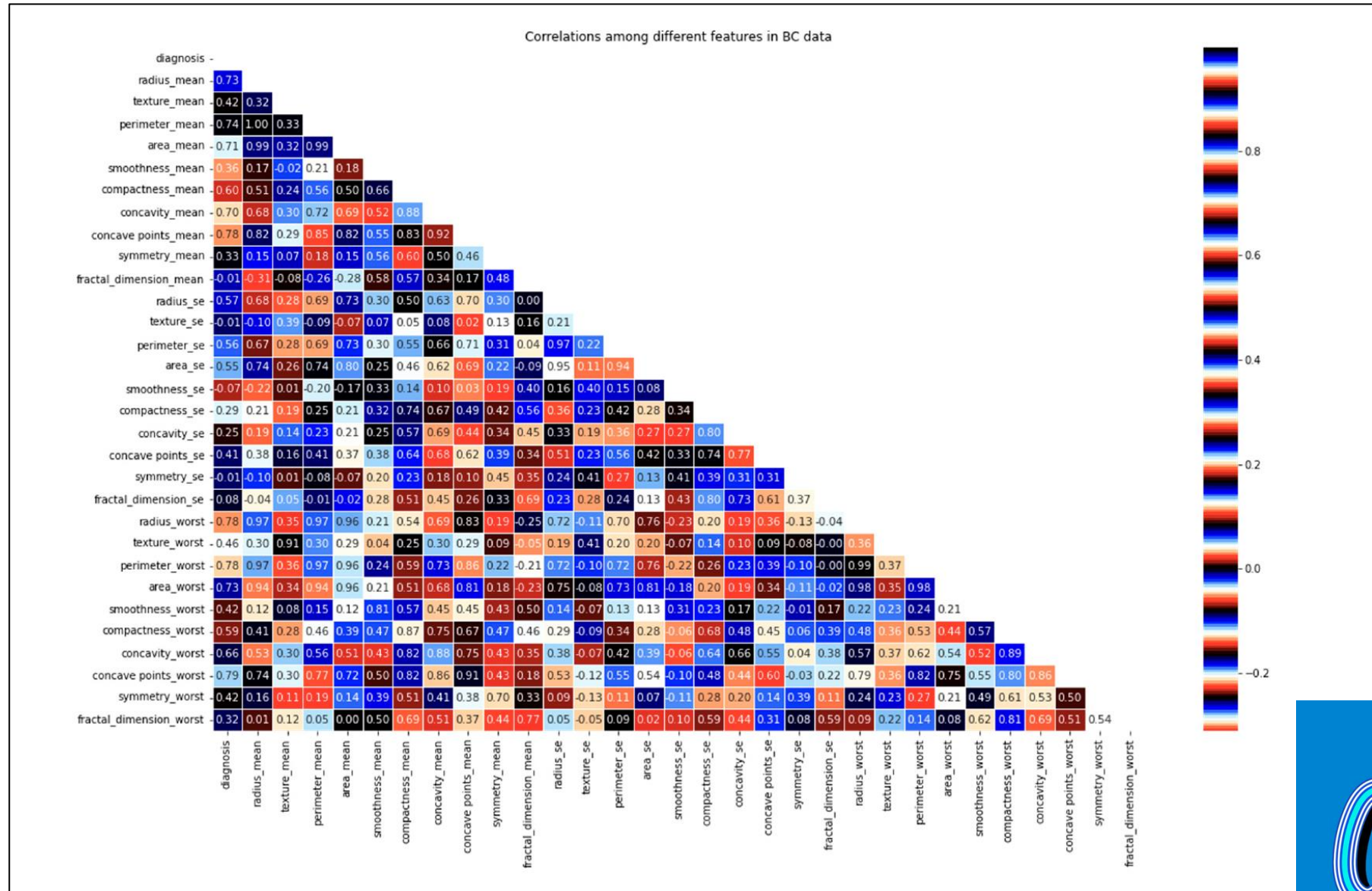
- Radius
- Area
- Perimeter



Heatmap for the Correlation among Different Features

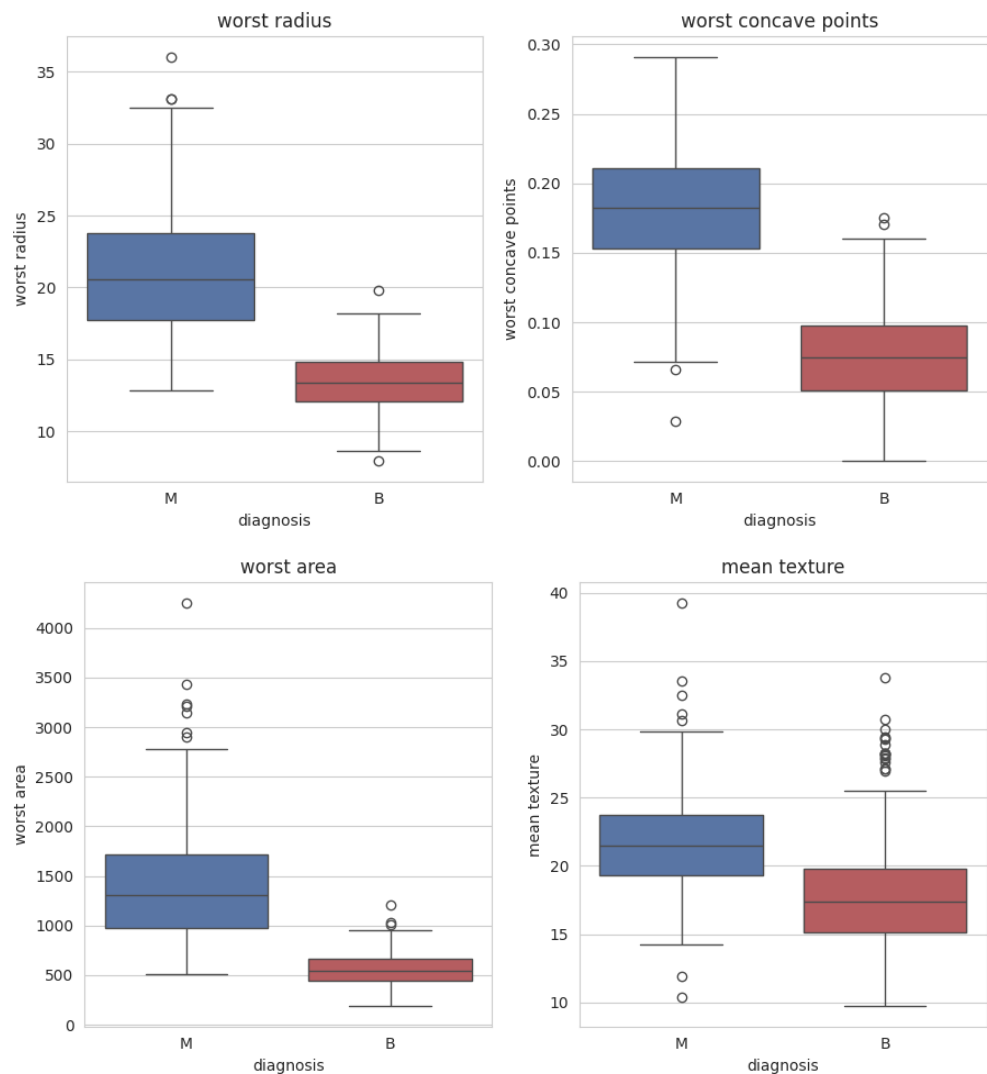
- To determine **how closely they are** (redundant). Values:
 - > 0 indicate a positive relationship
 - ~ 0 implies weak or relationship
- Features of the “mean”, “SE” and “Worst” versions **underline the same measurement**.
 - For example, “mean radius vs. worst radius” **are > 0.95 correlation** with each other.





Cool

Key Features by Class & Outliers Check for Malignancy



	Feature	Outlier count	Outlier %
13	area error	65	11.4
10	radius error	38	6.7
12	perimeter error	38	6.7
23	worst area	35	6.2
14	smoothness error	30	5.3
19	fractal dimension error	28	4.9
15	compactness error	28	4.9
18	symmetry error	27	4.7
3	mean area	25	4.4
29	worst fractal dimension	24	4.2

Quiz Question?

What does the feature “Worst Area” mean ?



Super!

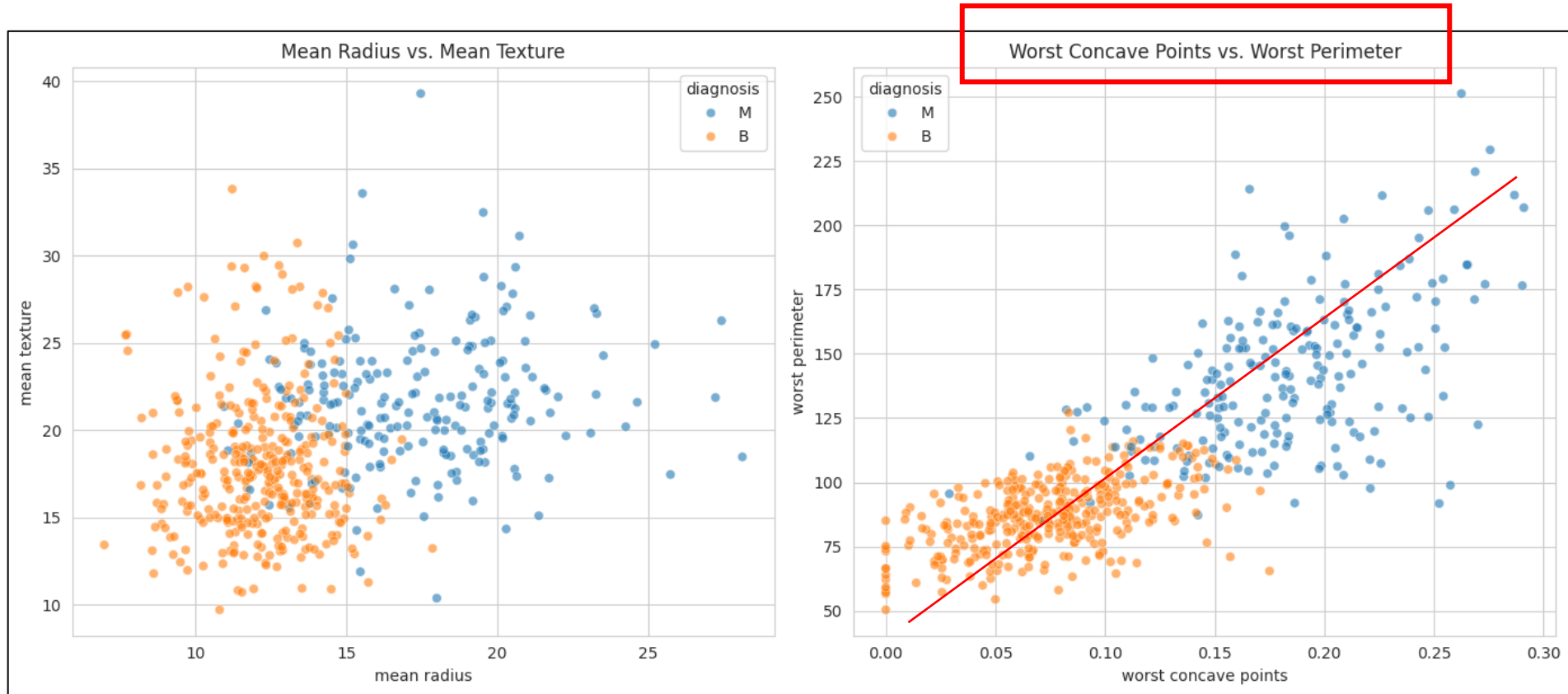
Across all the **nuclei**, we have **3 measurements**:

Mean: the average value across all nuclei

SE (standard error): how much the values vary between nuclei

Worst: the **average of the three largest** (most extreme) values
among all nuclei

Features Correlation with the Target Class



Non-Linear & Linear Multi Feature Relationships

Clean separation between the class with model overlap for the: Shape-regularity and Size features

2- Data Processing

- These features are **real-valued parameters** that are assed **using digital image of breast mass aspirate**.
- The depict features of the cell nuclei that are present in the image.
- From these features, namely radius, perimiter, area, smoothness, texture, compactness, concavity, symmetry, concave points, and fractional dimensions.
- In the Data set, there are **no missing values**.

Shape: (569, 32)																						
	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points	mean symmetry	mean fractal dimension	radius error	texture error	perimeter error	area error	smoothness error	compactness error	concavity error	concave points error	symmetry error	fractal dimension error	worst radius t	
0	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	0.14710	0.2419	0.07871	1.0950	0.9053	8.589	153.40	0.006399	0.04904	0.05373	0.01587	0.03003	0.006193	25.38	
1	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	0.07017	0.1812	0.05667	0.5435	0.7339	3.398	74.08	0.005225	0.01308	0.01860	0.01340	0.01389	0.003532	24.99	
2	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	0.12790	0.2069	0.05999	0.7456	0.7869	4.585	94.03	0.006150	0.04006	0.03832	0.02058	0.02250	0.004571	23.57	
3	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	0.10520	0.2597	0.09744	0.4956	1.1560	3.445	27.23	0.009110	0.07458	0.05661	0.01867	0.05963	0.009208	14.91	
4	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	0.10430	0.1809	0.05883	0.7572	0.7813	5.438	94.44	0.011490	0.02461	0.05688	0.01885	0.01756	0.005115	22.54	

3- Features Engineering

- **New features** were engineered, motivated by the **hypothesis that malignant tumors are larger**, more irregular, and more heterogeneous than benign ones.

`worst_mean_radius_ratio`

Worst-to-mean radius: growth irregularity

`concavity_compactness_ratio`

Concavity relative to compactness: boundary roughness

`radius_concavepoints_interaction`

Radius × concave points: combined signal

`worst_mean_texture_ratio`

Worst-to-mean texture: texture heterogeneity

30

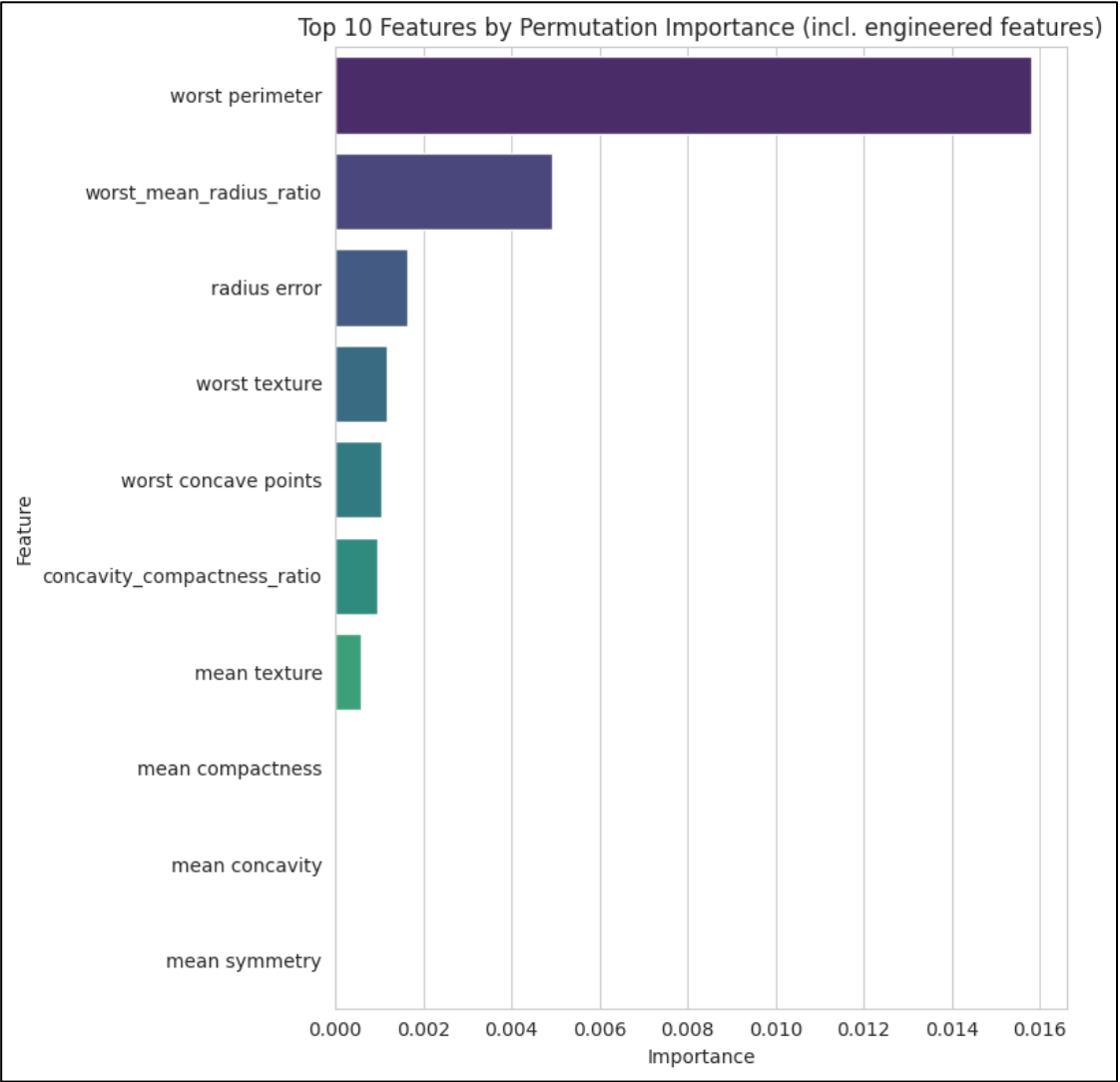
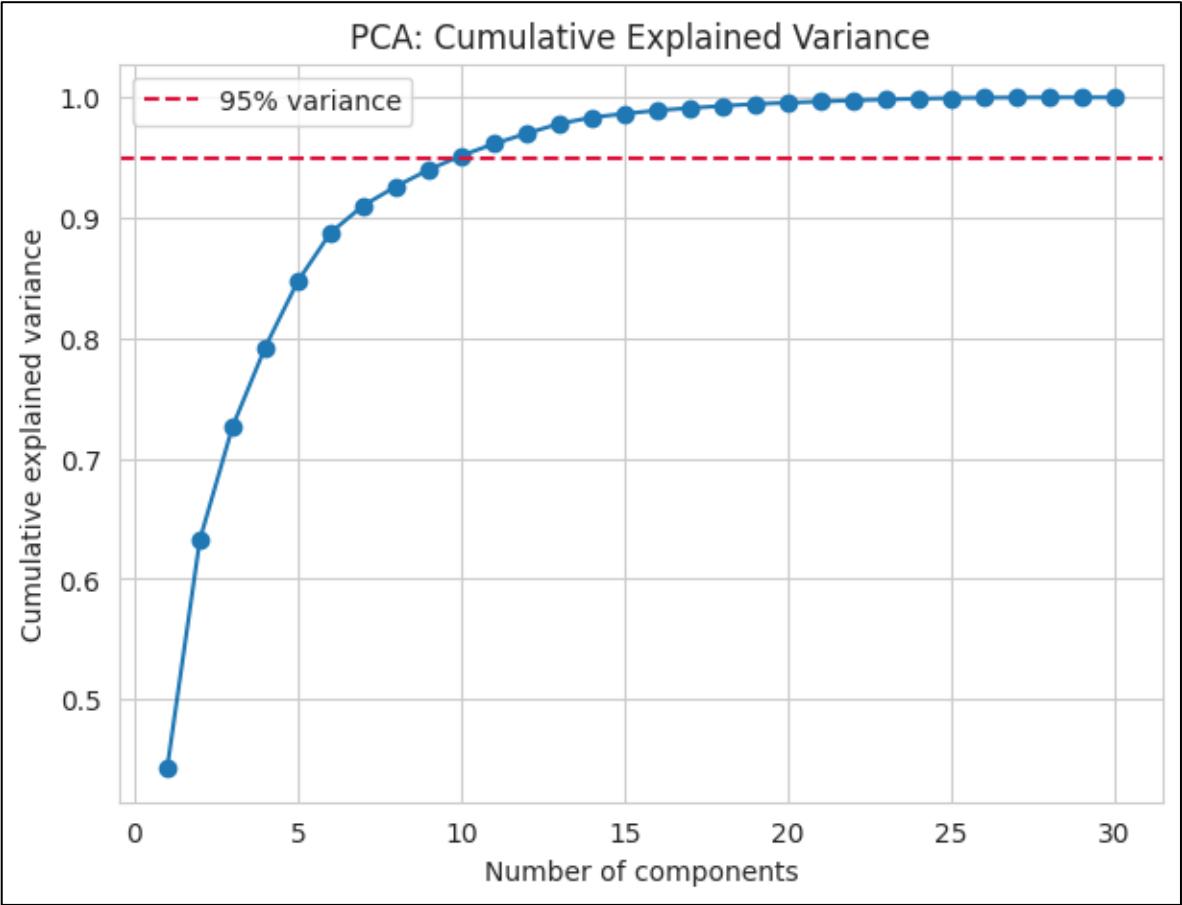
Raw features

27

Final feature set
(23 base + 4 engineered)

Features dropped: area error · mean area · mean perimeter · mean radius · perimeter error · worst area · worst radius

Features Importance & PCA



Baseline Model (Logistic Regression)

- After feature engineering, we have 569 samples and 27 continuous numeric, moderately correlated features.
- This is a binary classification (Malignant = 1 vs. Benign = 0).
- **Train/test split:** stratified 80/20 split (455 train / 114 test)
- **Naïve baseline:** majority class and random predictor
- **Cross validation:** 5-fold CV on the training
- **Tuning:** using GridSearchCV
- **Evaluation:** precision, accuracy, sensitivity, specificity, and F1-score, alongside ROC-AUC

Predicted true M

Missing M

Predicted true B

$$\text{Precision} = \frac{TP}{(FP + TP)}$$

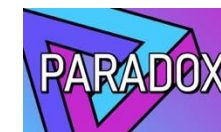
$$\text{Accuracy} = \frac{TP + TN}{(TP + TN + FN + FP)}$$

$$\text{Sensitivity} = \frac{TP}{(FN + TP)}$$

$$\text{Specificity} = \frac{TN}{(FN + TN)} \times 100\%$$

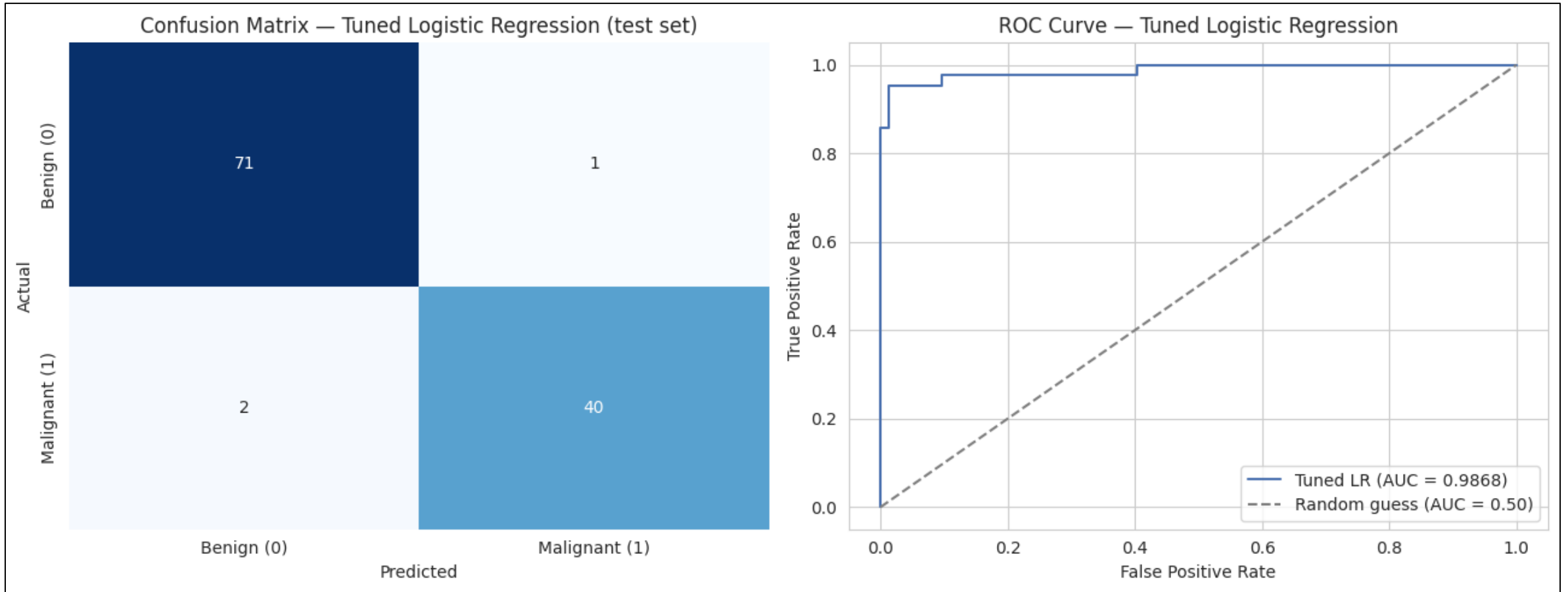
$$\text{F1-Score} = 2 \times \frac{(\text{Recall} \times \text{Precision})}{(\text{Recall} + \text{Precision})}$$

	Accuracy	Precision	Recall	F1	Specificity	ROC-AUC
Naive - Majority Class	0.6316	0.0000	0.0000	0.0000	NaN	NaN
Naive - Random (stratified)	0.5965	0.4474	0.4048	0.4250	NaN	NaN
Baseline LR (default)	0.9737	1.0000	0.9286	0.9630	1.0000	0.9940
Tuned LR	0.9737	0.9756	0.9524	0.9639	0.9861	0.9868

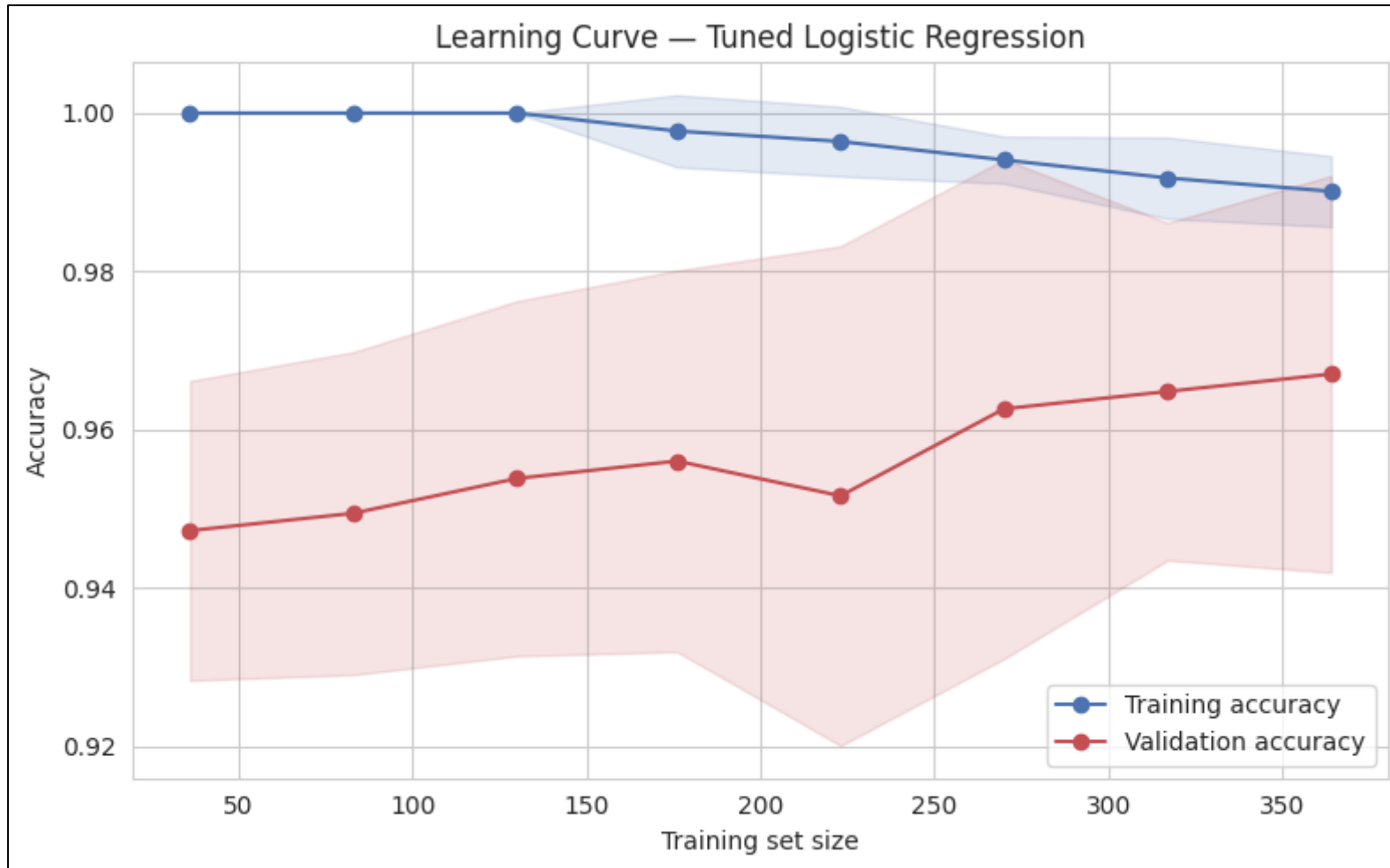


Sensitivity: Catching one more true Malignant case at the cost of one extra false positive

Error Analysis- Confusion Matrix & ROC Curve for Tuned LR



Tuned Logistic Regression- Fitting Check



1) Results for (LR)

```
Logistic Regression
Accuracy   : 0.9474
Precision  : 0.9737 (positive class = malignant)
Recall     : 0.881 (Sensitivity for malignant)
Specificity: 0.9861 (Specificity for benign)
```

Confusion Matrix:

```
[[71  1]
 [ 5 37]]
```

Classification Report:

	precision	recall	f1-score	support
benign (0)	0.93	0.99	0.96	72
malignant (1)	0.97	0.88	0.93	42
accuracy			0.95	114
macro avg	0.95	0.93	0.94	114
weighted avg	0.95	0.95	0.95	114

```
(0.9473684210526315, 0.9736842105263158)
```

2) Results for (KNN, SVM)

KNN (k=7)

Accuracy : 0.9561

Precision : 1.0 (positive class = malignant)

Recall : 0.881 (Sensitivity for malignant)

Specificity: 1.0 (Specificity for benign)

Confusion Matrix:

```
[[72  0]
 [ 5 37]]
```

Classification Report:

	precision	recall	f1-score	support
benign (0)	0.94	1.00	0.97	72
malignant (1)	1.00	0.88	0.94	42
accuracy			0.96	114
macro avg	0.97	0.94	0.95	114
weighted avg	0.96	0.96	0.96	114

(0.956140350877193, 1.0)

SVM (RBF)

Accuracy : 0.9649

Precision : 0.975 (positive class = malignant)

Recall : 0.9286 (Sensitivity for malignant)

Specificity: 0.9861 (Specificity for benign)

Confusion Matrix:

```
[[71  1]
 [ 3 39]]
```

Classification Report:

	precision	recall	f1-score	support
benign (0)	0.96	0.99	0.97	72
malignant (1)	0.97	0.93	0.95	42
accuracy			0.96	114
macro avg	0.97	0.96	0.96	114
weighted avg	0.97	0.96	0.96	114

(0.9649122807017544, 0.975)

3) Results for (DT, RF)

Decision Tree (max_depth=5)

Accuracy : 0.9035
Precision : 0.9429 (positive class = malignant)
Recall : 0.7857 (Sensitivity for malignant)
Specificity: 0.9722 (Specificity for benign)

Confusion Matrix:

```
[[70  2]
 [ 9 33]]
```

Classification Report:

	precision	recall	f1-score	support
benign (0)	0.89	0.97	0.93	72
malignant (1)	0.94	0.79	0.86	42
accuracy			0.90	114
macro avg	0.91	0.88	0.89	114
weighted avg	0.91	0.90	0.90	114

(0.9035087719298246, 0.9428571428571428)

Random Forest (n=300)

Accuracy : 0.9737
Precision : 1.0 (positive class = malignant)
Recall : 0.9286 (Sensitivity for malignant)
Specificity: 1.0 (Specificity for benign)

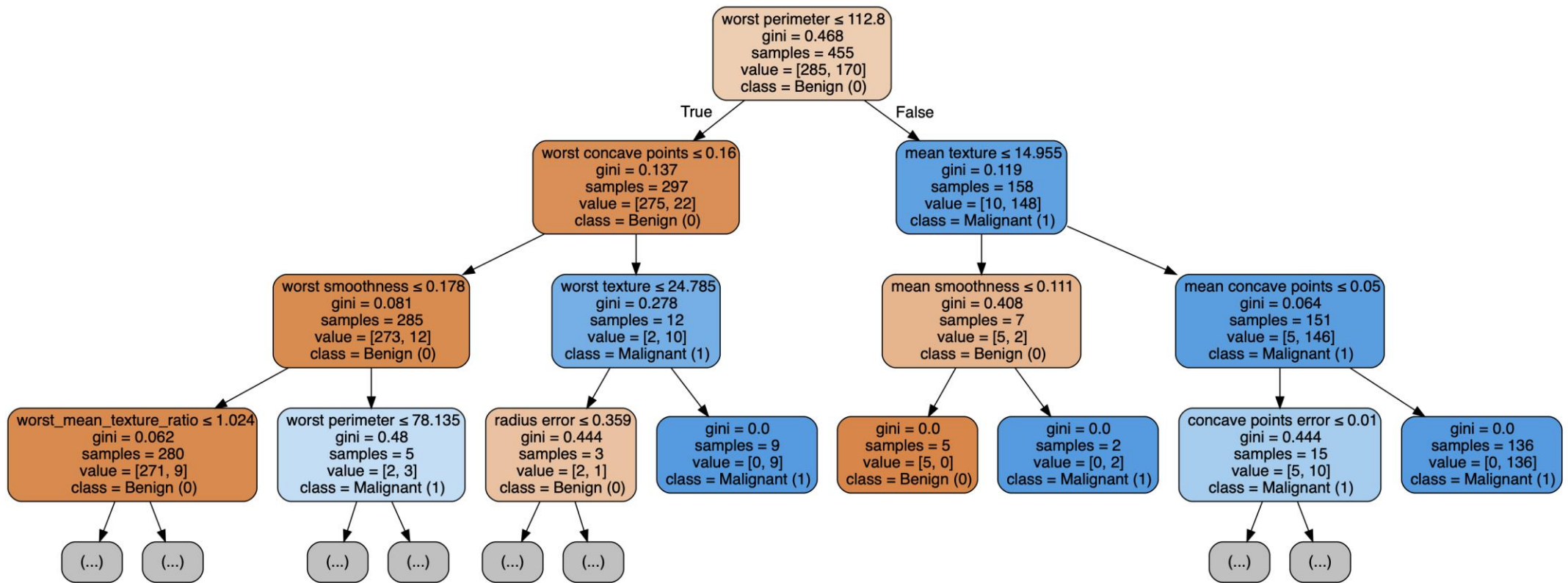
Confusion Matrix:

```
[[72  0]
 [ 3 39]]
```

Classification Report:

	precision	recall	f1-score	support
benign (0)	0.96	1.00	0.98	72
malignant (1)	1.00	0.93	0.96	42
accuracy			0.97	114
macro avg	0.98	0.96	0.97	114
weighted avg	0.97	0.97	0.97	114

(0.9736842105263158, 1.0)

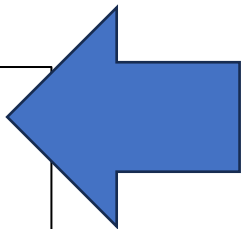


Cool

3) Ensemble Learning (Voting, Bagging, Boosting, Stacking)

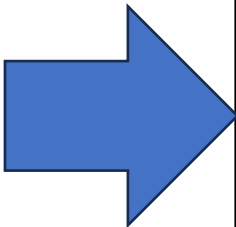
Top 5 Models by Accuracy:

2- Ensemble Learning (RF baseline)



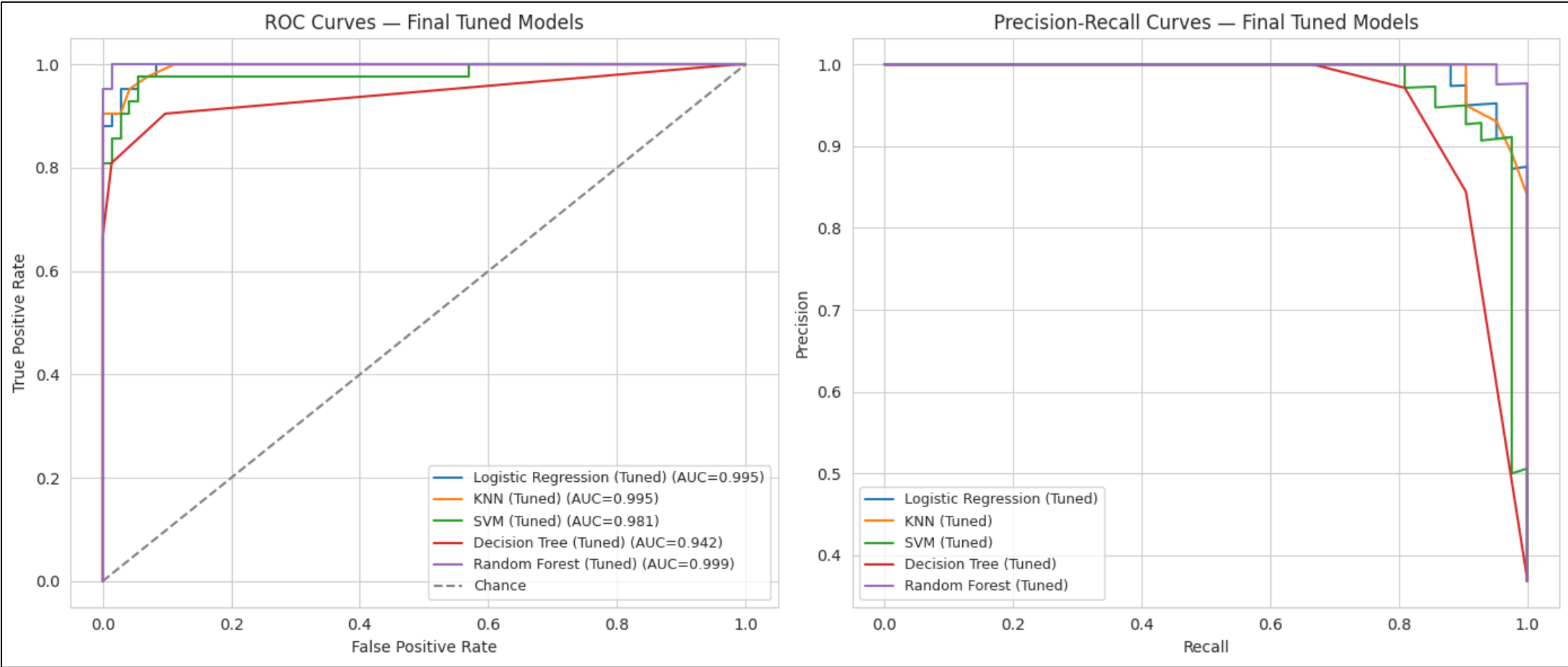
	Accuracy	Precision	Recall	Specificity
Ensemble - Voting (Soft)	0.9825	1.0	0.9524	1.0
Ensemble - Stacking	0.9825	1.0	0.9524	1.0
Ensemble - Gradient Boosting	0.9825	1.0	0.9524	1.0
Ensemble - AdaBoost	0.9825	1.0	0.9524	1.0
Random Forest (Tuned)	0.9737	1.0	0.9286	1.0

1- Hyperparameter Tuning

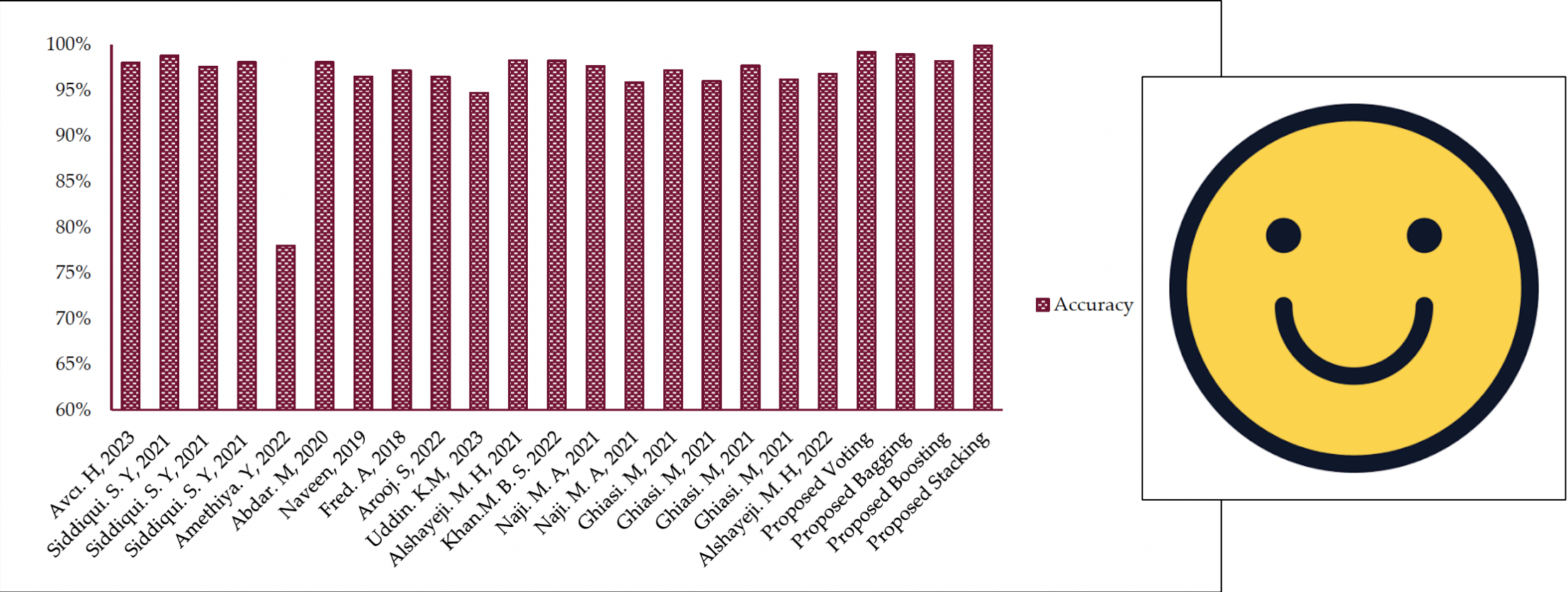


	Untuned Accuracy	Tuned Accuracy	Improvement
Logistic Regression	0.9474	0.9561	0.0088
KNN	0.9561	0.9561	0.0000
SVM	NaN	0.9474	NaN
Decision Tree	0.9035	0.9211	0.0175
Random Forest	0.9737	0.9737	0.0000

Models Comparison

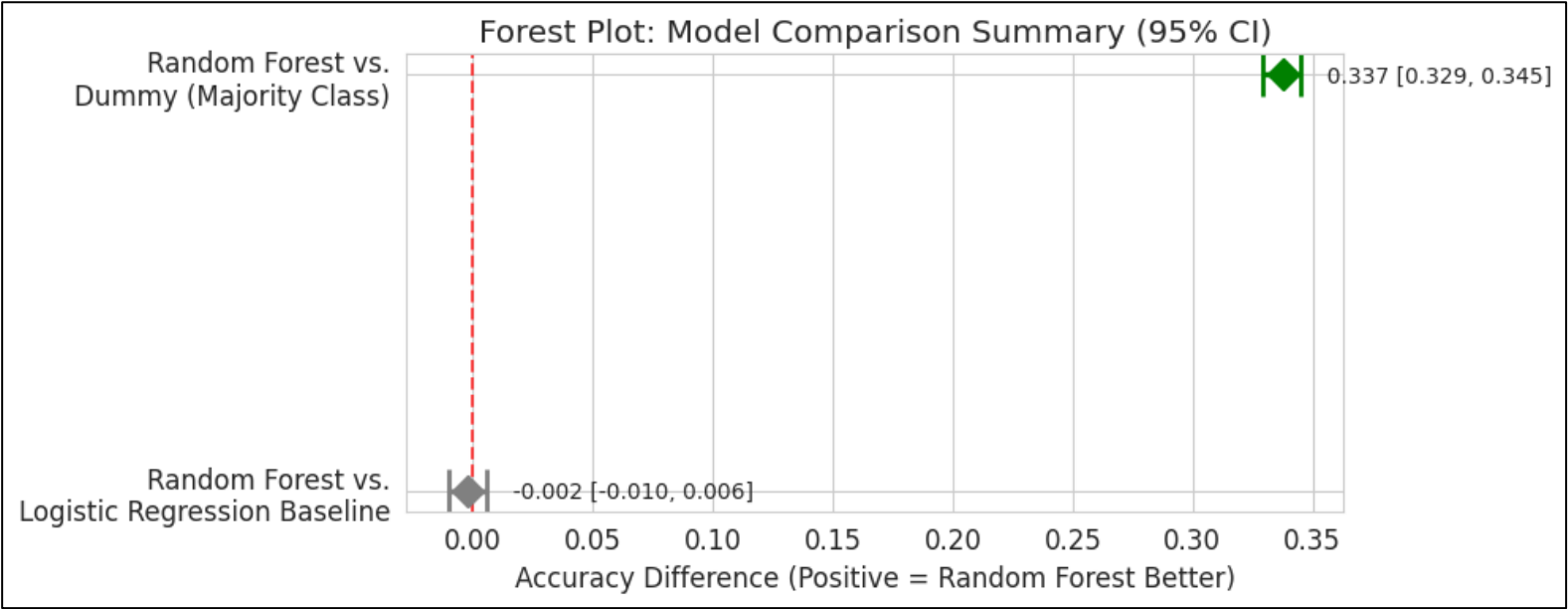


BEFORE you come at me, here's what the industry says



Statistical Significance and CIs

- I compared the “Tuned Random Forest” the best-performing complex model against two reference models:
 - “tuned logistic regression baseline”
 - “naïve majority-class dummy classifier”
- Methods used:
 - Paired t-test:** the differences are normal distribution
 - Wilcoxon:** robust on skewed distribution
 - McNemar’s test:** where the two models disagree on individual predictions
 - Cohen’s D test:** measures the size of a difference between two groups



=== Statistical Significance Summary: Model Comparisons ===							
	Comparison	Mean Diff (Accuracy)	95% CI	p-value (t-test)	p-value (Wilcoxon)	p-value (McNemar)	Cohen's d (alpha=0.05)?
0	Random Forest vs. LR Baseline	-0.0024	[-0.0100, 0.0058]	0.577222	0.393864	0.838256	-0.1047
1	Random Forest vs. Dummy (Majority Class)	0.3375	[0.3294, 0.3451]	4.11e-36	-	-	-

95% CI: Confident the true accuracy difference

MLP (Multi-Layer Perceptron) as an Alternative Model

- **Unlike Random Forest**, a small MLP significantly outperformed the tuned Logistic Regression baseline (+0.82pp mean CV accuracy, $p = 0.0015$, Cohen's $d = 0.65$)
- Suggesting deeper non-linear signal exists in the data than tree-based ensembles captured.

4/4		0s	35ms/step
4/4		0s	13ms/step
Accuracy:	0.9737		
Precision:	0.9756		
Sensitivity:	0.9524		
Specificity:	0.9861		
ROC-AUC:	0.9970		

Conclusion & Reflection

- Hypothesis is answered.
- Strengths, limitations, and future work:
 - Small (569-sample), single-institution dataset
- Biases: FNA from one institution
- Key takeaways.
 - Statistical testing can change a project's conclusion

Topic/Outcome	External Validation	Multi-Center Data	Deep Learning	XAI Integration	Clinical Deployment
Benign vs. Malignant Classification	2	GAP	12	8	3
Feature Importance & Interpretability	1	GAP	4	10	2
Privacy-Preserving Diagnosis	GAP	2	3	4	1
Cross-Dataset Generalization	3	1	5	2	1
Survival & Prognosis Prediction	1	GAP	2	1	GAP



Broader Societal Impact

This kind of model could help clinicians triage FNA samples faster and more consistently, especially valuable where specialist pathologists are scarce. But over-reliance without clinician oversight, or deployment on a meaningfully different patient population without revalidation, risks silently degraded, unevenly distributed accuracy.



Thank You!

Questions?